

Evaluation of Autoencoders as a dimensionality reduction technique for high-dimensional financial data

Pavel Klapatsiuk a12247946@unet.univie.ac.at
Maryia Bashmakova a12245851@unet.univie.ac.at

University of Vienna, Austria

Thesis proposal

In that seminar paper we propose the use of neural networks with autoencoder architecture as a dimensionality reduction technique and comparing it to Principal component analysis. Autoencoder models might have an advantage over PCA as they allow to perform a non-linear transformation of the input high-dimensional signal and our goal for that thesis is to check the feasibility of that assumption by applying both approaches to a 5-year historical dataset of S&P 500 companies stock prices [1]. We are going to compare these dimension reduction techniques both as standalone approaches via comparing reconstruction error and entropy, and as tools for feature preparation for two prediction techniques: a static factor model and an LSTM neural network.

Usage of Autoencoders as a dimensionality-reduction tool is a widely researched topic and several autoencoder-based methods can be found in [2], [3].

References

1. Nugent, C.: S&P 500 Stock Data. Kaggle dataset, historical stock data for all current S&P 500 companies (updated 2018). <https://www.kaggle.com/datasets/camnugent/sandp500> :contentReference[oaicite:0]index=0
2. Gunduz, H. An efficient stock market prediction model using hybrid feature reduction method based on variational autoencoders and recursive feature elimination. *Financial Innovation* 7(1), 28 (2021). <https://doi.org/10.1186/s40854-021-00243-3>
3. Cortés, D.G., Onieva, E., López, I.P., Trinchera, L., Wu, J.: Autoencoder-Enhanced Clustering: A Dimensionality Reduction Approach to Financial Time Series. *IEEE Access* 12, 16999–17009 (2024). <https://doi.org/10.1109/ACCESS.2024.3359413>